



SINGAPORE  
MANAGEMENT  
UNIVERSITY

**ISSS602: Data Analytics Lab**

**Assignment 1: Show Me the Numbers**

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**NOR HENDRA BIN ABDUL RAHMAN  
STUDENT ID: 01505038**

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## 1. Objective

The objective of this report is to quantify the drivers of HappyRides hourly demand using rigorously prepared data and SAS Viya analyses (EDA and confirmatory tests), and translate the findings into concise, evidence-based recommendations.

## 2. Data Pre-Look

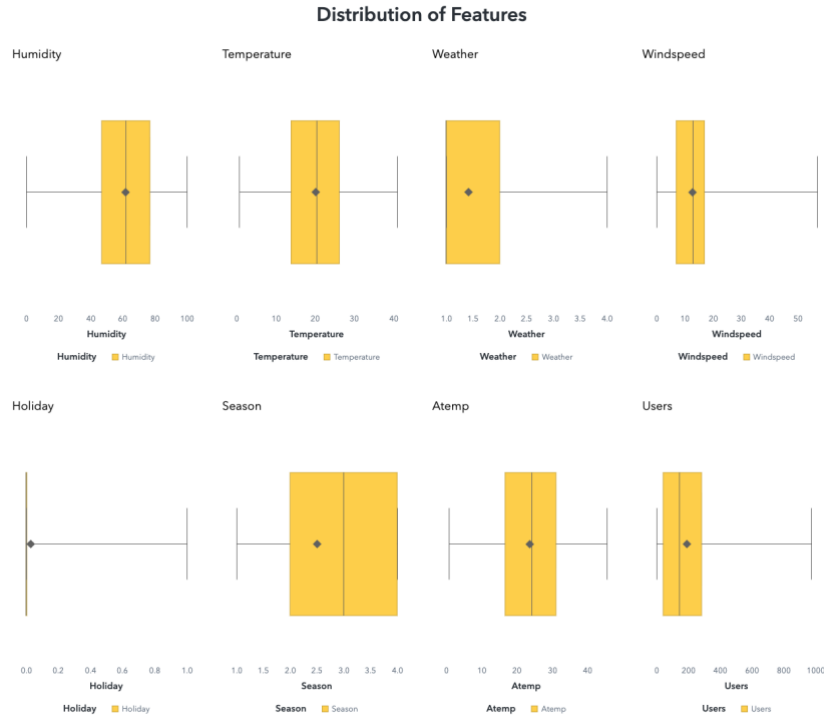


Figure 1: Distribution of Features

From the boxplot pre-look, the *Users* distribution is clearly right-skewed with a long upper whisker, indicating occasional high-demand hours that pull the mean above the median. *Humidity* is bounded yet concentrated around the mid-to-high range (roughly 50–75%), with a right tail suggesting more humid outliers. *Temperature* and *atemp* have similar centers and spreads with mild right tails, visually signalling redundancy/collinearity—only one should be used at a time in modelling. *Windspeed* shows a pronounced right-skew with sporadic high-wind outliers. For the categorical fields, *Weather* is dominated by levels 1–2 with very few cases at 3–4 (imbalance), *Season* is uneven but usable (fewer observations at level 1), and *Holiday* is almost entirely non-holiday (rare positives). Overall, the shapes and whiskers in these boxplots already foreshadow non-normality and outliers across several variables, motivating robust summaries (median/IQR), careful handling of rare categories, and avoidance of jointly including temp and atemp in the same model.

### 3. Data Preparation

#### 3.1. Pre-Checks

Pre-Checks — Summary

| Check                               | Value  | Status | Note |
|-------------------------------------|--------|--------|------|
| Rows (total in CSV)                 | 10,886 |        |      |
| Rows with any missing value         | 0      | OK     | —    |
| Duplicate datetime keys (distinct)  | 0      | OK     | —    |
| Rows involved in duplicate keys     | 0      | OK     | —    |
| Extra rows beyond first occurrences | 0      | OK     | —    |
| casual + registered != count (rows) | 0      | OK     | —    |

Figure 2: Summary of Pre-Checks

The raw file contains 10,886 rows and passed all front-door validations: no rows with missing values, no duplicate datetime keys (hence no duplicate-involved or “extra” rows to resolve), and the accounting identity casual + registered = count holds for every record. Together, these results indicate a clean, uniquely keyed hour-level table suitable for downstream EDA and inference without corrective preprocessing; any remaining quality work can focus on temporal completeness (gaps), outliers, and distributional assumptions rather than basic data hygiene.

#### 3.2. Temperature Anomaly

```
data atemp_temp_scan;
  set _d_sorted;
  /* Differences */
  delta_atemp_temp = atemp - temp;
  abs_delta        = abs(delta_atemp_temp);
  flag_large_gap    = (abs_delta > &GAP_THRESH);

  /* Stable-compare atemp using rounding */
  atemp_r = round(atemp, &ATEMP_ROUND);

  /* Track consecutive-hour runs with same rounded atemp */
  retain prev_dt prev_atemp_r run_len run_id;
  if _n_ = 1 then do;
    run_id = 1;
    run_len = 1;
  end;
  else if intck('hour', prev_dt, dt) = 1 and atemp_r = prev_atemp_r then do
    ;
    run_len + 1;
  end;
  else do;
    run_id + 1;
    run_len = 1;
  end;

  flag_const_run = (run_len >= &RUN_MIN);

  /* carry forward */
  prev_dt      = dt;
  prev_atemp_r = atemp_r;

  format delta_atemp_temp abs_delta 6.2;
run;
```

Figure 3: Code snippet for atemp anomaly (Assign1/1\_DataPreparation/2\_TempAnomaly.sas)

Top 20 hours by |atemp - temp|

| Obs | dt                 | temp  | atemp | delta_atemp_temp | abs_delta | weather | humidity | windspeed | count |
|-----|--------------------|-------|-------|------------------|-----------|---------|----------|-----------|-------|
| 1   | 17AUG2024:14:00:00 | 35.26 | 12.12 | -23.14           | 23.14     | 1       | 34       | 27.9993   | 328   |
| 2   | 17AUG2024:15:00:00 | 35.26 | 12.12 | -23.14           | 23.14     | 1       | 30       | 31.0009   | 383   |
| 3   | 17AUG2024:13:00:00 | 34.44 | 12.12 | -22.32           | 22.32     | 1       | 36       | 26.0027   | 357   |
| 4   | 17AUG2024:16:00:00 | 34.44 | 12.12 | -22.32           | 22.32     | 2       | 32       | 30.0026   | 488   |
| 5   | 17AUG2024:12:00:00 | 33.62 | 12.12 | -21.50           | 21.50     | 1       | 41       | 15.0013   | 365   |
| 6   | 17AUG2024:17:00:00 | 33.62 | 12.12 | -21.50           | 21.50     | 1       | 36       | 22.0028   | 791   |
| 7   | 17AUG2024:18:00:00 | 33.62 | 12.12 | -21.50           | 21.50     | 2       | 38       | 16.9979   | 669   |
| 8   | 17AUG2024:11:00:00 | 31.16 | 12.12 | -19.04           | 19.04     | 1       | 52       | 19.0012   | 303   |
| 9   | 17AUG2024:10:00:00 | 30.34 | 12.12 | -18.22           | 18.22     | 1       | 55       | 11.0014   | 231   |
| 10  | 17AUG2024:19:00:00 | 30.34 | 12.12 | -18.22           | 18.22     | 2       | 55       | 26.0027   | 491   |
| 11  | 17AUG2024:20:00:00 | 29.52 | 12.12 | -17.40           | 17.40     | 2       | 58       | 15.0013   | 359   |
| 12  | 17AUG2024:09:00:00 | 28.70 | 12.12 | -16.58           | 16.58     | 1       | 58       | 7.0015    | 378   |
| 13  | 17AUG2024:00:00:00 | 27.88 | 12.12 | -15.76           | 15.76     | 1       | 57       | 11.0014   | 88    |
| 14  | 17AUG2024:08:00:00 | 27.88 | 12.12 | -15.76           | 15.76     | 1       | 65       | 8.9981    | 668   |
| 15  | 17AUG2024:21:00:00 | 27.88 | 12.12 | -15.76           | 15.76     | 2       | 69       | 19.9995   | 255   |
| 16  | 17AUG2024:01:00:00 | 27.06 | 12.12 | -14.94           | 14.94     | 1       | 65       | 7.0015    | 54    |
| 17  | 17AUG2024:02:00:00 | 27.06 | 12.12 | -14.94           | 14.94     | 1       | 61       | 8.9981    | 19    |
| 18  | 17AUG2024:22:00:00 | 27.06 | 12.12 | -14.94           | 14.94     | 3       | 83       | 12.9980   | 213   |
| 19  | 17AUG2024:03:00:00 | 26.24 | 12.12 | -14.12           | 14.12     | 1       | 65       | 7.0015    | 6     |
| 20  | 17AUG2024:04:00:00 | 26.24 | 12.12 | -14.12           | 14.12     | 1       | 73       | 11.0014   | 9     |

Figure 4: Results of atemp anomaly checks

Using a 10 °C gap threshold and ≥6-hour constant-value rule, the scan surfaced systematic issues with atemp: on 17 Aug 2024, atemp is frozen at 12.12 °C for 24 consecutive hours, producing extreme gaps versus temp (e.g., change up to 23.14 °C) and multiple hours flagged as anomalies; similar constant-atemp runs (10–25 h) appear in other periods as well. A collinearity snapshot shows  $r(\text{temp}, \text{atemp})=0.98495$ ,  $r^2=0.970$  and an implied  $\text{VIF} \approx 33.5$  if both are modeled together, it is beyond acceptable limits. Taken together, these results indicate:

- i. atemp contains anomalies that would skew data
- ii. even when not anomalous, atemp is redundant with temp.

For analysis, *atemp* will therefore be excluded or imputed in flagged runs, and choose *temp* to avoid instability from multicollinearity.

### 3.3. Date Time Checks

```

42  /* 4) Assemble compact list of missing HH per day */
43  ⊖ data _miss_list;
44      set _cal_flag;
45      where is_observed = 0;
46      hour_txt = put(hour, z2.);
47      keep date hour_txt;
48      format date ddmmyy10.;
49  run;
50
51  proc sort data=_miss_list; by date hour_txt; run;
52
53  ⊖ data missing_hours_by_day;
54      length missing_hours $300;
55      retain missing_hours n_missing;
56      format date ddmmyy10.;
57      keep date n_missing missing_hours;
58      set _miss_list;
59      by date;
60      if first.date then do; missing_hours=''; n_missing=0; end;
61      missing_hours = catx(' ', missing_hours, hour_txt);
62      n_missing + 1;
63      if last.date then output;
64  run;

```

Figure 5: Code snippet of missing hours (Assign1/1\_DataPreparation/3\_DateTimeCheck.SAS)

A date and hour calendar join shows systematic gaps, with recurring full-day outages (24 missing hours) on the 20th to 31st across many 2023–2024 months and occasional single-hour holes, consistent with source-level truncation rather than random loss. To neutralize these gaps, all summaries use Average Users calculated over observed hours only, and daily comparisons retain all days while relying on the observed-hours denominator to handle uneven coverage. Statistical inference uses tests robust to unequal sample sizes, such as the Welch t test. The coverage denominator is reported for several views and will be added wherever missing so that demand rates, seasonal or weekday effects, and peak analyses remain comparable without downward bias from missing hours.

### 3.4. Weather/Season/Holiday Check

```

14  /* Frequency tables for categorical variables */
15  proc sql;
16      /* SEASON */
17      create table _freq_season as
18      select 'season' as var length=8,
19             case when missing(season) then 'MISSING'
20                  else strip(put(season, best.))
21             end as level length=12,
22             count(*) as N
23      from casuser.Bike_sharing_dataset
24      group by calculated level;
25
26      /* HOLIDAY */
27      create table _freq_holiday as
28      select 'holiday' as var length=8,
29             case when missing(holiday) then 'MISSING'
30                  else strip(put(holiday, best.))
31             end as level length=12,
32             count(*) as N
33      from casuser.Bike_sharing_dataset
34      group by calculated level;
35
36      /* WEATHER */
37      create table _freq_weather as
38      select 'weather' as var length=8,
39             case when missing(weather) then 'MISSING'
40                  else strip(put(weather, best.))
41             end as level length=12,
42             count(*) as N
43      from casuser.Bike_sharing_dataset
44      group by calculated level;
45  quit;

```

Figure 6: Code snippet of weather, season and holiday checks.  
(Assign1/1\_DataPreparation/4\_WeatherSeasonHolidayCheck.SAS)

Season is well balanced across the four codes (about one quarter each), so seasonal comparisons are statistically stable. Weather is concentrated in fair conditions: level 1 accounts for 66.1% of hours, level 2 for 26.0%, level 3 for 7.9%, and level 4 appears only once in the entire sample (one hour on 09/01/2024), which makes extreme weather effectively absent. Holidays are rare at 2.9% of hours overall (2023: 2.7%, 2024: 3.1%), with 6-7 holiday days per year. These profiles imply that holiday should be treated as a rare-event flag, weather level 4 should be grouped with adjacent adverse categories or excluded from inference, and chi-square style tests should verify minimum expected counts or use grouping to maintain validity.

### 3.5. Season Sanity Check

```
4      /* 1) Parse dt, derive month/year, expected season by meteorological rule */
5  ⊖ data _season_check;
6      set casuser.Bike_sharing_dataset(keep='datetime'n season);
7      length season_name season_exp_name $8 month_name $9;
8      dt = input(strip('datetime'n), anydtdtm.);
9      if missing(dt) then delete;
10
11     date = datepart(dt);
12     year = year(date);
13     month = month(date);
14
15     /* Expected season by calendar month: 1=Spring,2=Summer,3=Fall,4=Winter */
16     if month in (3,4,5) then season_expected = 1; /* Mar-May */
17     else if month in (6,7,8) then season_expected = 2; /* Jun-Aug */
18     else if month in (9,10,11) then season_expected = 3; /* Sep-Nov */
19     else season_expected = 4; /* Dec-Feb */
20
```

Figure 7: Code snippet for season sanity checks. (Assign1/1\_DataPreparation/5\_SeasonSanityCheck.sas)

The dataset's season field aligns with calendar quarters rather than meteorological seasons. Counts place season=1 in January-March, season=2 in April-June, season=3 in July-September, and season=4 in October-December, yielding 100% mismatches in months outside March, June, September, and December when compared to the meteorological rule. Detected change points occur at 00:00 on April 1, July 1, October 1, and January 1, not on March 1, June 1, September 1, and December 1. This indicates a labelling or encoding shift and will be treated as a quarter code and relabel before interpretation.

### 3.6. Target Leakage of Users Count

```
19     /* 1) Correlations: r(count, casual) and r(count, registered), with r^2 */
20 ⊖ proc corr data=casuser.Bike_sharing_dataset pearson outp=_corr noprint;
21     var 'count'n casual registered;
22 run;
23
24 ⊖ data count_corr_summary;
25     set _corr;
26     if _TYPE_='CORR' and lowercase(_NAME_='count' then do;
27         r_count_casual = casual;
28         r2_count_casual = r_count_casual**2;
29         r_count_registered = registered;
30         r2_count_registered = r_count_registered**2;
31         output;
32     end;
33     keep r_count_casual r2_count_casual r_count_registered r2_count_registered;
34 run;
35
36 title "Leakage signal: Correlations with target";
37 ⊖ proc print data=count_corr_summary noobs label;
38     label r_count_casual = "r(count, casual)"
39           r2_count_casual = "r^2(count, casual)"
40           r_count_registered = "r(count, registered)"
41           r2_count_registered = "r^2(count, registered)";
42     format r_count_ r2_count_ 8.3;
43 run;
```

Figure 8: Code snippet for target leakage of users. (Assign1/1\_DataPreparation/6\_TargetLeakage.sas)



#### Leakage signal: Correlations with target

| r(count, casual) | r^2(count, casual) | r(count, registered) | r^2(count, registered) |
|------------------|--------------------|----------------------|------------------------|
| 0.690            | 0.477              | 0.971                | 0.943                  |

#### Target distribution: zeros/low counts + quantiles

| Rows  | Zeros<br>(n) | Zeros<br>(%) | Count<br>≤ 3<br>(n) | Count<br>≤ 3<br>(%) | Mean   | SD     | Max    | Min  | P50 | P90 | P95 | P99 |
|-------|--------------|--------------|---------------------|---------------------|--------|--------|--------|------|-----|-----|-----|-----|
| 10886 | 0            | 0.00%        | 381                 | 3.50%               | 191.57 | 181.14 | 977.00 | 1.00 | 145 | 452 | 564 | 775 |

Figure 9: Results for Target Leakage

The identity check confirms no mismatches in  $\text{count} = \text{casual} + \text{registered}$ . Correlations show count is dominated by the registered series ( $r = 0.971$ ,  $r^2 = 0.943$ ) and only moderately aligned with casual ( $r = 0.690$ ,  $r^2 = 0.477$ ). The target has no zeros and a pronounced right tail (P50 = 145 users/hour, P90 = 452, P95 = 564, P99 = 775; max = 977), with only 3.5 percent of rows at three users or fewer. Nighttime hours concentrate the very low counts, while commute and evening periods carry the high medians. Composition shares are stable and skewed toward registered riders (median registered share  $\approx 0.86$ ; casual  $\approx 0.14$ ). Methodologically, casual and registered will be treated as components of the target to avoid using them as predictors with count to prevent leakage.

### 3.7. Weekend Flag

```

6  ⊖ data _weekend_flags;
7      set casuser.Bike_sharing_dataset(keep='datetime'n);
8      dt  = input(strip('datetime'n), anydtdtm.);
9      if missing(dt) then delete;
10     date = datepart(dt);
11     dow  = weekday(date);
12     length day_name $9;
13     day_name = put(date, downname.);
14     is_weekend = (dow in (1,7));
15     label is_weekend = 'Weekend? (Sat/Sun=1)';
16     dow          = 'Day-of-week (1=Sun .. 7=Sat)';
17     format date ddmmyy10. dt datetime19.;
18     keep dt date dow day_name is_weekend;
19     run;

```

Figure 10: Code snippet for weekend flag. (Assign1/1\_DataPreparation/7\_WeekendFlag.sas)

Weekend flag summary (Sat/Sun=1)

| Weekend? | n    | pct    |
|----------|------|--------|
| 1        | 3113 | 28.60% |
| 0        | 7773 | 71.40% |

By day-of-week (1=Sun ... 7=Sat)

| Day-of-week (1=Sun .. 7=Sat) | day_name  | n    | Rows flagged as weekend |
|------------------------------|-----------|------|-------------------------|
| 1                            | Sunday    | 1584 | 1584                    |
| 2                            | Monday    | 1579 | 0                       |
| 3                            | Tuesday   | 1551 | 0                       |
| 4                            | Wednesday | 1539 | 0                       |
| 5                            | Thursday  | 1551 | 0                       |
| 6                            | Friday    | 1553 | 0                       |
| 7                            | Saturday  | 1529 | 1529                    |

Figure 11: Results of weekend flag script.

Weekend hours (Saturday and Sunday) account for 28.6% of observations ( $n = 3,113$ ), while weekdays account for 71.4% ( $n = 7,773$ ). Counts by day confirm correct tagging with all Sunday rows ( $n = 1,584$ ) and Saturday rows ( $n = 1,529$ ) are flagged as weekend, with zero weekend flags on Monday to Friday. This validates the Sat/Sun weekend definition for subsequent weekday-weekend comparisons.

### 3.8. Data Transformation

```

18 /* 1) Weekend flags */
19 data bike_step2_weekend;
20 set bike_step1_time;
21 dow = weekday(date);
22 length day_name $9;
23 day_name = put(date, downname.);
24 is_weekend = (dow in (1,7));
25
26 label dow = 'Day-of-week (1=Sun..7=Sat)';
27 day_name = 'Day name';
28 is_weekend = 'Weekend? (Sat/Sun=1)';
29 format date ddmmyy10. dt datetime19.;
30 run;
31
32 /* 2) Weekend split sanity */
33 proc sql noprint;
34 select count(*) into :_total_rows from bike_step2_weekend;
35 quit;
36
37 proc sql;
38 create table weekend_summary as
39 select is_weekend,
40 count(*) as n,
41 calculated n / &_total_rows as pct format=percent8.2
42 from bike_step2_weekend
43 group by is_weekend
44 order by is_weekend desc;
45 quit;
46
47 proc print data=weekend_summary noobs label;
48 var is_weekend n pct;
49 label is_weekend='Weekend?';
50 run;
51
52 /* 3) Weather */
53 data bike_step3_weather;
54 set bike_step2_weekend;
55
56 length weather_grp 8 weather_grp_cat $8;
57 select (weather);
58 when (1) do; weather_grp=1; weather_grp_cat='Good'; end;
59 when (2) do; weather_grp=2; weather_grp_cat='Mist'; end;
60 when (3,4) do; weather_grp=3; weather_grp_cat='Adverse'; end;
61 otherwise do; weather_grp=.; weather_grp_cat='Unknown'; end;
62 end;
63
64 label weather_grp = 'Weather group (1=Good,2=Mist,3=Adverse)';
65 weather_grp_cat = 'Weather group (label)';
66
67 run;
68
69 /* 3B) Sanity checks */
70 title "Weather group distribution (numeric 1/2/3)";
71 proc freq data=bike_step3_weather;
72 tables weather_grp / missing;
73 run;
74
75 title "Weather group (character labels)";
76 proc freq data=bike_step3_weather;
77 tables weather_grp_cat / missing;
78 run;
79
80 /* 4) Drop ATEMP */
81 data bike_step4_noatemp;
82 set bike_step3_weather;
83 drop atemp;
84 run;
85
86 title "Schema after dropping ATEMP";
87 proc contents data=bike_step4_noatemp varnum; run;
88 title;

```

Figure 12: Data Transformation code. (Assign1/1\_DataPreparation/8\_FinalDataOutput.sas)

This block creates analysis-ready features and removes a problematic field. The timestamp stored as text is parsed into a numeric datetime (dt) and split into date and hour to support hourly models, calendar grouping, and exact joins on a date-by-hour key. A weekend flag (Saturday or Sunday) is derived from the day-of-week so that weekday versus weekend demand can be compared directly and summarized consistently. Weather codes are collapsed into three groups which are Good = 1, Mist = 2, Adverse = 3 or 4, to address extreme sparsity in level 4, improve expected counts for contingency tests, and preserve the fair-versus-adverse contrast needed for ANOVA or rank-based comparisons. The variable atemp is then removed because earlier checks found blockwise constant runs and very high collinearity with temp, which would inflate standard errors and create instability if both were modelled together.

### 3.9. Binning of Variables

*Refer to folder: Assign1/2\_ExploreVisualize*

**Temperature** was binned in 3-degree bands from 0-3 up to 39-42 to capture smooth curvature and the comfort band visible in the charts, including the local maximum at 30-33 and softening above the high-30s. These narrow bands let the mean profile trace the rise and easing without overfitting to noise.

**Humidity** was binned in 10-point bands to align with its broad, monotonic effect and to support the 60% reference (Eren & Emre Uz, 2020) that separates high-throughput hours from suppressed demand in the report.

**Windspeed** was binned in 5-unit steps to distinguish the “calm” dip from the broad mid-range plateau around 10-30 seen in the usage curve.

**Season** was handled as categorical levels to compare demand across quarters in a way that matches the strong temperature shifts by season.

**A weekend type flag** enables clean stratification for hour-of-day comparisons and paired tests.

Finally, **peak-hour bins** group the day into Morning, Midday, Evening, and Off-peak so that the recurring commute band and the broader evening band show up as level differences, which ANOVA and Tukey procedures can test directly. This design also matches the heatmap’s two dominant windows without fragmenting the sample across 24 separate hours.

These choices mirror the empirical patterns and tests presented in the report: the temperature peak at 30-33, the humidity drop, the wind mid-range plateau, the seasonal ordering driven mainly by temperature, and the AM/PM concentration of use.

## 4. Data Analyses

Analyses were conducted in SAS Viya. Hypothesis testing used Levene's test for homogeneity; when variances were unequal, Welch's ANOVA (Satterthwaite df) was applied with Tukey-Kramer post-hoc comparisons. A 95% confidence level ( $\alpha = 0.05$ ) was used; results with  $p < 0.05$  were deemed statistically significant and the null hypothesis rejected.

Table analyses used Pearson  $\chi^2$  with observed/expected counts, row/column percentages, and measures of association (e.g., Cramér's V) to test independence and quantify effect size; expected counts check  $\chi^2$  assumptions, while percentages standardize for marginal imbalances to aid interpretation.

### 4.1. Insight 1: Weather is a clear demand driver

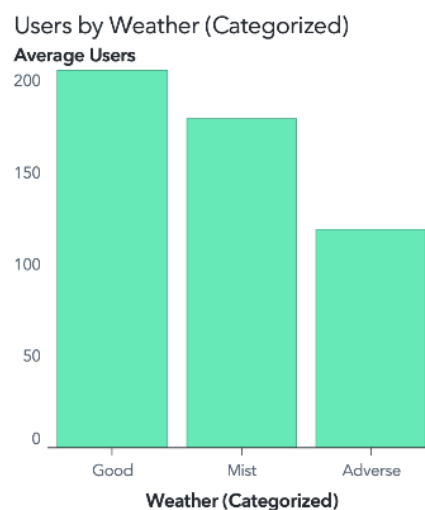


Figure 13: Average Users vs Weather(Categorized)

| Levene's Test for Homogeneity of count Variance<br>ANOVA of Squared Deviations from Group Means |       |                |             |         |        |
|-------------------------------------------------------------------------------------------------|-------|----------------|-------------|---------|--------|
| Source                                                                                          | DF    | Sum of Squares | Mean Square | F Value | Pr > F |
| weather_grp_cat                                                                                 | 2     | 2.586E11       | 1.293E11    | 37.77   | <.0001 |
| Error                                                                                           | 10883 | 3.726E13       | 3.4239E9    |         |        |

| Welch's ANOVA for count |        |         |        |
|-------------------------|--------|---------|--------|
| Source                  | DF     | F Value | Pr > F |
| weather_grp_cat         | 2.0000 | 140.97  | <.0001 |
| Error                   | 2452.5 |         |        |

Figure 14: Levene Test and Welch's Anova

**H<sub>0</sub>:** The mean hourly users are equal across weather categories ( $\mu_{\text{Good}} = \mu_{\text{Mist}} = \mu_{\text{Adverse}}$ ).

**H<sub>1</sub>:** At least one weather category mean differs.

Average users are highest on Good weather days, then Mist, and lowest under Adverse. Welch's ANOVA (used after Levene indicated unequal variances) shows a significant weather effect,  $F = 140.97$ ,  $p < .0001$ , with small model fit ( $R^2 \approx 0.018$ ); Tukey-Kramer indicates every pair differs (all  $p < .0001$ ). Group means (users/hour)

are ≈205 (Good), 179 (Mist), 119 (Adverse), underscoring a material gap between favorable and adverse conditions.

#### 4.2. Insight 2: Temperature Distribution differs by Weather

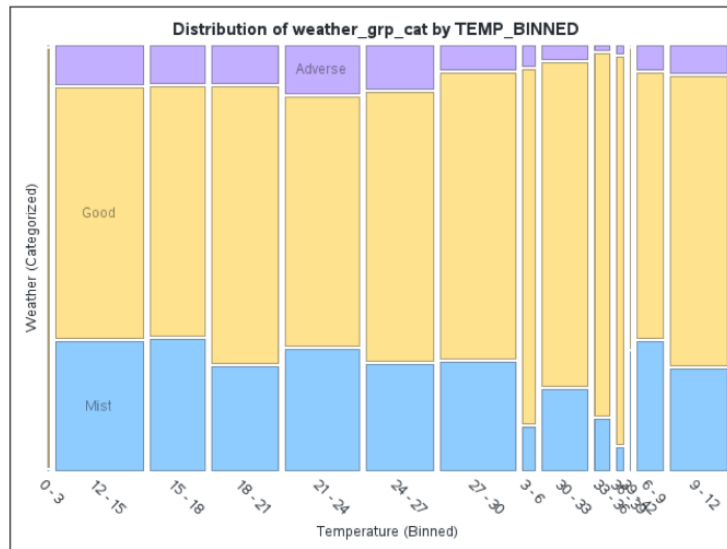


Figure 15: Temperature by Weather(Categorized)

| Statistic                   | DF | Value    | Prob   |
|-----------------------------|----|----------|--------|
| Chi-Square                  | 26 | 301.7831 | <.0001 |
| Likelihood Ratio Chi-Square | 26 | 338.3871 | <.0001 |
| Mantel-Haenszel Chi-Square  | 1  | 2.0098   | 0.1563 |
| Phi Coefficient             |    | 0.1665   |        |
| Contingency Coefficient     |    | 0.1642   |        |
| Cramer's V                  |    | 0.1177   |        |

Figure 16: Statistics for Temperature and Weather

**H<sub>0</sub>:** Temperature bin and weather category are independent (no association).

**H<sub>1</sub>:** Temperature bin and weather category are associated (not independent).

Temperature spreads vary by weather: Good clusters around mid-temperatures with tighter IQRs; Mist is broader; Adverse is sparser and more variable. The contingency test TEMP\_BINNED by Weather is significant ( $\chi^2(26)=301.78$ ,  $p<.0001$ ), but Cramér's  $V=0.118$  indicates a weak association which is useful for nuance, not a dominant driver. The observed-vs-expected table supports mid-temperature enrichment under Good and relative under-representation for Adverse in several bins.

### 4.3. Insight 3: Humidity Profile differs by Weather

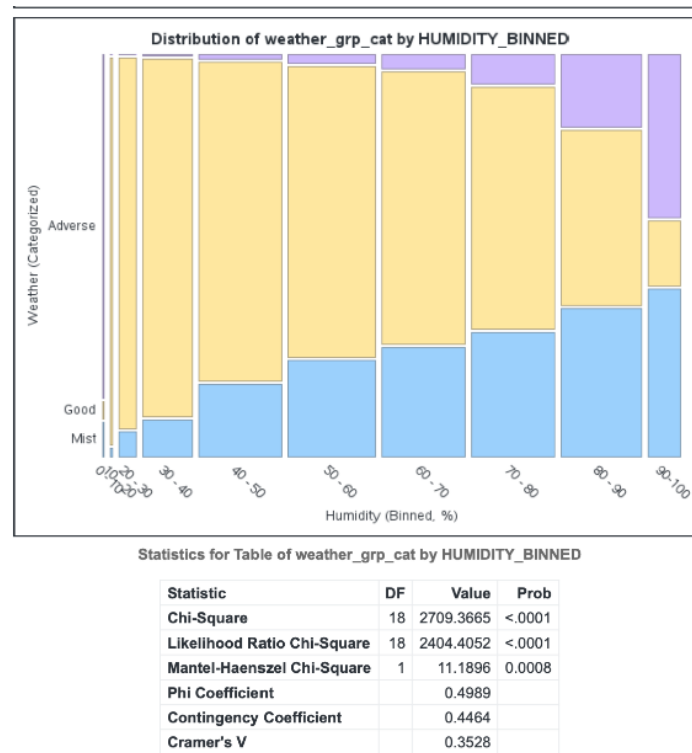


Figure 17: Distribution of Weather Category by Humidity Binned, and its Statistics

**H<sub>0</sub>:** Humidity bin and weather category are independent.

**H<sub>1</sub>:** Humidity bin and weather category are associated.

The humidity profile differs by weather: Good weather concentrates in mid-humidity ( $\approx 40\text{--}70\%$  RH) with tighter IQRs; Mist shifts slightly higher and broader; Adverse is up-weighted at high humidity ( $80\text{--}100\%$  RH). Pearson's  $\chi^2$  confirms moderate association ( $\chi^2(18)=2709.37$ ,  $p<.0001$ ; Cramér's  $V=0.353$ ), with  $N=10,886$ . Observed vs expected counts show mid-band enrichment for Good and over-representation of Adverse in the upper bins.

#### 4.4. Insight 4: User and Humidity Association

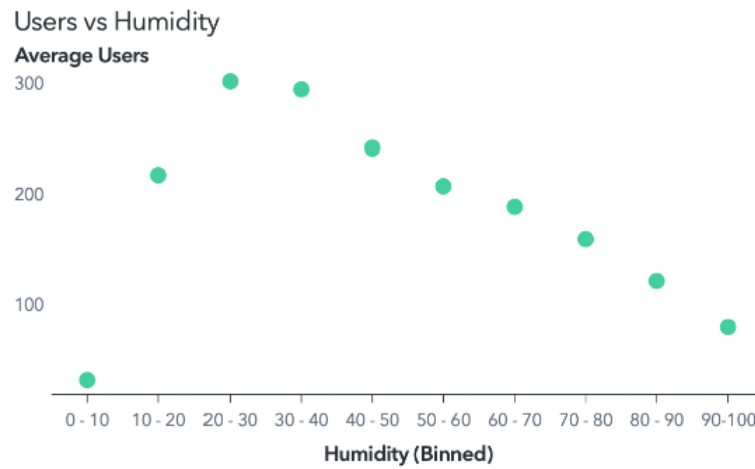


Figure 18: Dot Plot of Average Users vs Humidity

| Levene's Test for Homogeneity of count Variance<br>ANOVA of Squared Deviations from Group Means |       |                |             |         |        |
|-------------------------------------------------------------------------------------------------|-------|----------------|-------------|---------|--------|
| Source                                                                                          | DF    | Sum of Squares | Mean Square | F Value | Pr > F |
| HUMIDITY_BINNED                                                                                 | 9     | 6.475E11       | 7.194E10    | 24.36   | <.0001 |
| Error                                                                                           | 10876 | 3.212E13       | 2.9532E9    |         |        |

| Welch's ANOVA for count |        |         |        |
|-------------------------|--------|---------|--------|
| Source                  | DF     | F Value | Pr > F |
| HUMIDITY_BINNED         | 9.0000 | 240.89  | <.0001 |
| Error                   | 478.6  |         |        |

| Level of HUMIDITY_BINNED | N    | count      |            |
|--------------------------|------|------------|------------|
|                          |      | Mean       | Std Dev    |
| 0 - 10                   | 23   | 30.434783  | 28.169658  |
| 10 - 20                  | 45   | 217.066667 | 199.134765 |
| 20 - 30                  | 364  | 301.527473 | 186.947148 |
| 30 - 40                  | 1039 | 294.194418 | 202.543125 |
| 40 - 50                  | 1727 | 241.255356 | 180.635181 |
| 50 - 60                  | 1842 | 206.998371 | 181.832191 |
| 60 - 70                  | 1748 | 188.363844 | 181.257325 |
| 70 - 80                  | 1736 | 158.914747 | 162.881041 |
| 80 - 90                  | 1676 | 120.770286 | 138.099822 |
| 90-100                   | 686  | 78.860058  | 107.574537 |

Figure 19: Statistics from ANOVA test

**H<sub>0</sub>:** Mean hourly users are equal across humidity bins ( $\mu_{0-10} = \mu_{10-20} = \dots = \mu_{90-100}$ ).  
**H<sub>1</sub>:** At least one humidity-bin mean differs.

Users peak at 20–30% ( $\approx 301.5$  users/hr) and 30–40% ( $\approx 294.2$ ), then decline monotonically as humidity rises: 40–50%  $\approx 241.3 \rightarrow$  50–60%  $\approx 207.0 \rightarrow$  60–70%  $\approx 188.4 \rightarrow$  70–80%  $\approx 158.9 \rightarrow$  80–90%  $\approx 120.8 \rightarrow$  90–100%  $\approx 78.9$ . Variances are unequal (Levene  $p < .0001$ ), so Welch's ANOVA is primary ( $F(9, 478.6) = 240.89$ ,  $p < .0001$ ; classical one-way  $F(9, 10876) = 150.93$ ,  $p < .0001$ ;  $R^2 \approx 0.111$ ). Tukey–Kramer shows most adjacent drops are significant, while 20–30 vs 30–40 is not ( $p \approx 0.9995$ ), indicating a flat top before the downturn.

#### 4.5. Insight 5: Weather Composition Across Windspeeds

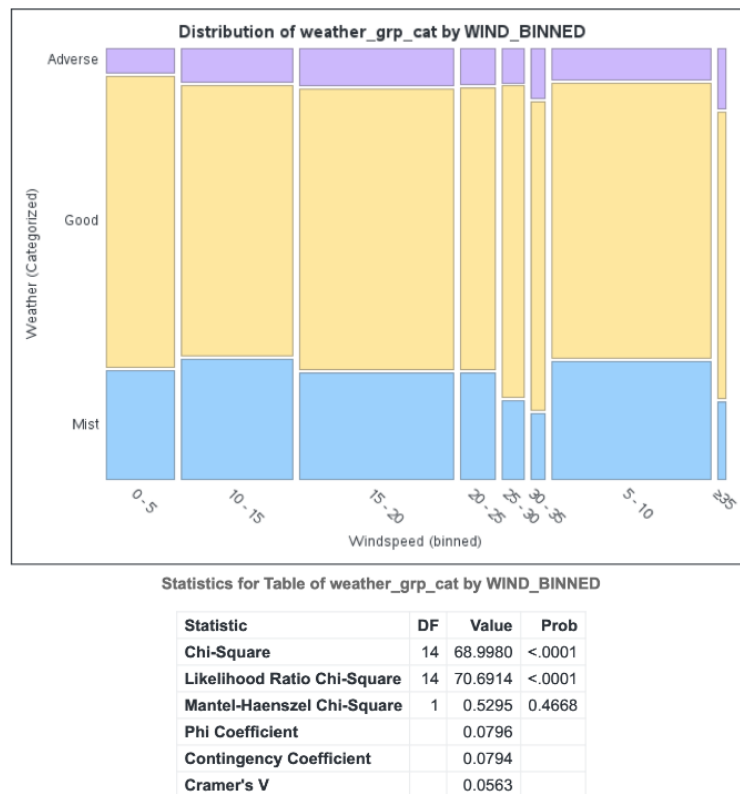


Figure 20: Distribution of Weather by Windspeed Binned, and its Statistics

$H_0$ : Windspeed bin and weather category are independent.

$H_1$ : Windspeed bin and weather category are associated.

Windspeed distributions are broadly similar across weather states, with Adverse showing a slightly higher median and wider spread. The contingency test is significant ( $\chi^2(14)=68.998$ ,  $p<.0001$ ), but the effect is weak (Cramér's  $V=0.056$ ). Interpretation: weather mix shifts a bit across wind bands, but not strongly enough to be a major driver.

#### 4.6. Insight 6: Mid-Range Plateau for Users vs Windspeed

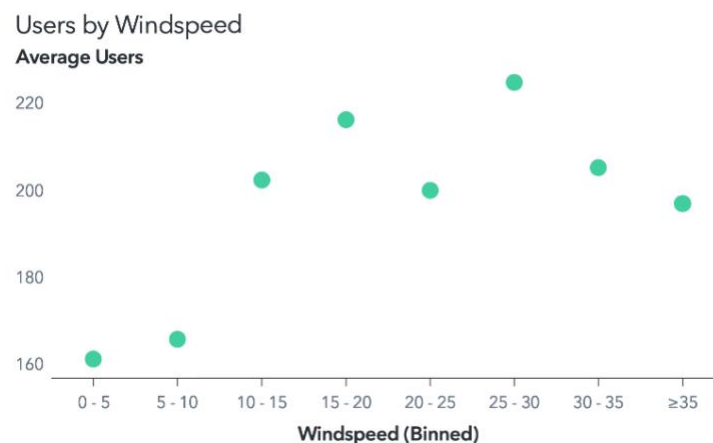


Figure 21: Average Users by Windspeed



| Levene's Test for Homogeneity of count Variance<br>ANOVA of Squared Deviations from Group Means |       |                |             |         |        |
|-------------------------------------------------------------------------------------------------|-------|----------------|-------------|---------|--------|
| Source                                                                                          | DF    | Sum of Squares | Mean Square | F Value | Pr > F |
| WIND_BINNED                                                                                     | 7     | 1.102E11       | 1.575E10    | 4.55    | <.0001 |
| Error                                                                                           | 10878 | 3.765E13       | 3.4614E9    |         |        |

| Welch's ANOVA for count |        |         |        |
|-------------------------|--------|---------|--------|
| Source                  | DF     | F Value | Pr > F |
| WIND_BINNED             | 7.0000 | 26.23   | <.0001 |
| Error                   | 1410.9 |         |        |

| Level of<br>WIND_BINNED | N    | count      |            |
|-------------------------|------|------------|------------|
|                         |      | Mean       | Std Dev    |
| 0 - 5                   | 1313 | 161.101295 | 172.221828 |
| 10 - 15                 | 2099 | 202.255836 | 186.745500 |
| 15 - 20                 | 2953 | 216.005418 | 189.354706 |
| 20 - 25                 | 646  | 199.880805 | 176.167599 |
| 25 - 30                 | 422  | 224.587678 | 182.423005 |
| 30 - 35                 | 280  | 205.103571 | 169.485623 |
| 5 - 10                  | 3026 | 165.660278 | 168.839759 |
| ≥35                     | 147  | 196.836735 | 187.995848 |

Figure 22: Users by Windspeed ANOVA statistics

**H<sub>0</sub>:** Mean hourly users are equal across windspeed bins ( $\mu_{0-5} = \mu_{5-10} = \dots = \mu_{\geq 35}$ ).

**H<sub>1</sub>:** At least one windspeed-bin mean differs.

Average users rise from calm (0–5: ~161; 5–10: ~166) to moderate winds (15–20: ~216; 25–30: ~225), then hover around a mid-range plateau (20–25: ~200; 30–35: ~205; ≥35: ~197). Variances are unequal (Levene  $p < .0001$ ), so the primary inference uses Welch's ANOVA,  $F(7, 1410.9) = 26.23$ ,  $p < .0001$  (classical one-way:  $F = 25.71$ ,  $R^2 \approx 0.016$ ). Tukey–Kramer shows the calm bins are lower than many mid-range bins, while several mid-range pairs are not statistically different, consistent with a broad plateau rather than a sharp optimum.

#### 4.7. Insight 7: Humidity and Windspeed Association, Temperature is Orthogonal

##### Weather & Meteorology Association

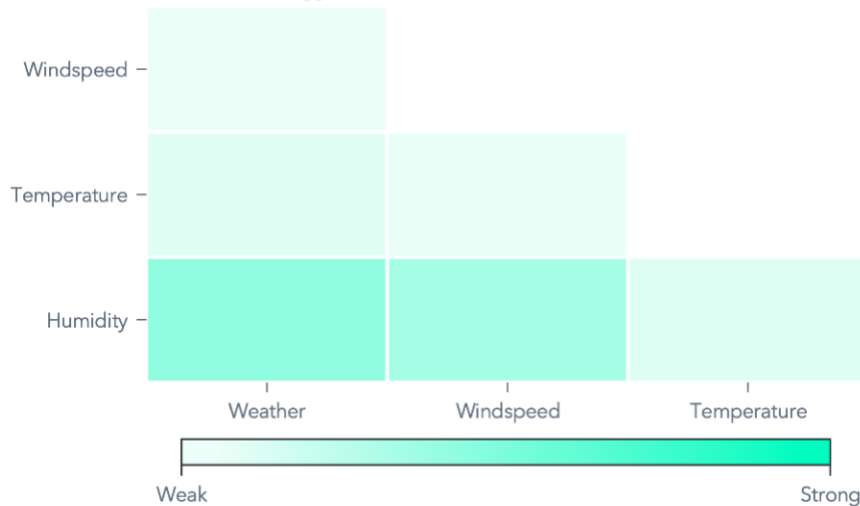


Figure 23: Mosaic plot for correlation

Having established each variable's association with users, pairwise correlations among temperature, windspeed, and humidity are examined to reveal interdependence that may confound effects. This step clarifies which meteorological factors move together versus independently, sharpening interpretation of drivers of demand and guarding against collinearity-driven misattribution.

3 Variables: temp humidity windspeed

| Pearson Correlation Coefficients, N = 10886<br>Prob >  r  under H0: Rho=0 |                    |                    |                    |
|---------------------------------------------------------------------------|--------------------|--------------------|--------------------|
|                                                                           | temp               | humidity           | windspeed          |
| temp<br>Temperature                                                       | 1.00000            | -0.06495<br><.0001 | -0.01785<br>0.0625 |
| humidity<br>Humidity                                                      | -0.06495<br><.0001 | 1.00000            | -0.31861<br><.0001 |
| windspeed<br>Windspeed                                                    | -0.01785<br>0.0625 | -0.31861<br><.0001 | 1.00000            |

| Spearman Correlation Coefficients, N = 10886<br>Prob >  r  under H0: Rho=0 |                    |                    |                    |
|----------------------------------------------------------------------------|--------------------|--------------------|--------------------|
|                                                                            | temp               | humidity           | windspeed          |
| temp<br>Temperature                                                        | 1.00000            | -0.04685<br><.0001 | -0.00554<br>0.5636 |
| humidity<br>Humidity                                                       | -0.04685<br><.0001 | 1.00000            | -0.32445<br><.0001 |
| windspeed<br>Windspeed                                                     | -0.00554<br>0.5636 | -0.32445<br><.0001 | 1.00000            |

Figure 24: Statistics from Correlation Analysis

$H_0$ :  $\rho = 0$  between each meteorological pair (Pearson/Spearman).

$H_1$ :  $\rho \neq 0$  between each pair.

Humidity and windspeed are moderately negatively correlated (Pearson  $r = -0.319$ ; Spearman  $\rho = -0.324$ , both  $p < .0001$ ), indicating windier periods tend to be drier. Temperature shows only a very weak negative tie to humidity (Pearson  $r = -0.065$ ; Spearman  $\rho = -0.047$ , both  $p < .0001$ ) and is near zero with windspeed (Pearson  $r = -0.018$ ,  $p = .0625$ ; Spearman  $\rho = -0.006$ ,  $p = .564$ ), implying temperature varies largely independently of wind in this dataset.

#### 4.8. Insight 8: Humidity as Primary Driver, Comfort Band for Temperature, Windspeed with Mid-Range Plateau

Given humidity's consistently strong signal with monotonic declines in users across bins, a moderate composition link to weather, and a negative coupling with windspeed. The analysis now drills deeper into humidity-specific effects. This includes testing practical thresholds (e.g., 60% RH) and a cutoff sweep to pinpoint actionable breakpoints that most affect demand.

```

1 ods noproctitle;
2 ods graphics / imagemap=on;
3
4 /* Macro to run normality, Welch two-sample t, and Wilcoxon for a cutoff */
5 %macro hum_cutoff(c=60);
6
7 title "Humidity threshold tests: cutoff=%c (group = <=cut vs >cut)";
8
9 /* 1) Normality within each group */
10 proc univariate data=CASUSER.HUMIDITY_THRESHOLD_LONG normal;
11   where cutoff=%c;
12   class group;
13   var count;
14   ods select TestsForNormality;
15 run;
16
17 /* 2) Two-sample t test (use Satterthwaite row for Welch) */
18 proc ttest data=CASUSER.HUMIDITY_THRESHOLD_LONG plots=none;
19   where cutoff=%c;
20   class group;
21   var count;
22   ods select Equality TTests ConLimits Statistics;
23 run;
24
25 /* 3) Nonparametric robustness (Wilcoxon rank-sum) */
26 proc npar1way data=CASUSER.HUMIDITY_THRESHOLD_LONG wilcoxon;
27   where cutoff=%c;
28   class group;
29   var count;
30   ods select WilcoxonTest;
31 run;
32
33 title;
34 %mend;
35
36 /* Run the finer grid below 60%, plus your higher checks */
37 %hum_cutoff(c=40);
38 %hum_cutoff(c=45);
39 %hum_cutoff(c=50);
40 %hum_cutoff(c=52);
41 %hum_cutoff(c=54);
42 %hum_cutoff(c=56);
43 %hum_cutoff(c=58);
44 %hum_cutoff(c=60);
45 %hum_cutoff(c=65);
46 %hum_cutoff(c=70);
47 %hum_cutoff(c=75);
48 --

```

```

1 data CASUSER.HAPPYRIDES_V9 FLAGGED;
2 set CASUSER.HAPPYRIDES_V9;
3 length HUMIDITY_60_FLAG $5;
4 if humidity <= 60 then HUMIDITY_60_FLAG='≤60%';
5 else HUMIDITY_60_FLAG='>60%';
6 run;

```

Figure 25: Code snippet for CDA prep (Assign1/3\_Confirmatory Data Analysis/5A\_HumiditySweepPrep.sas)

Above code chunk was used for the data preparation for this CDA for the Two-sample t Tests. This will find from our dataset which range users started falling off, and aims to check if it conforms with the research paper where users fell off after 60% Humidity.

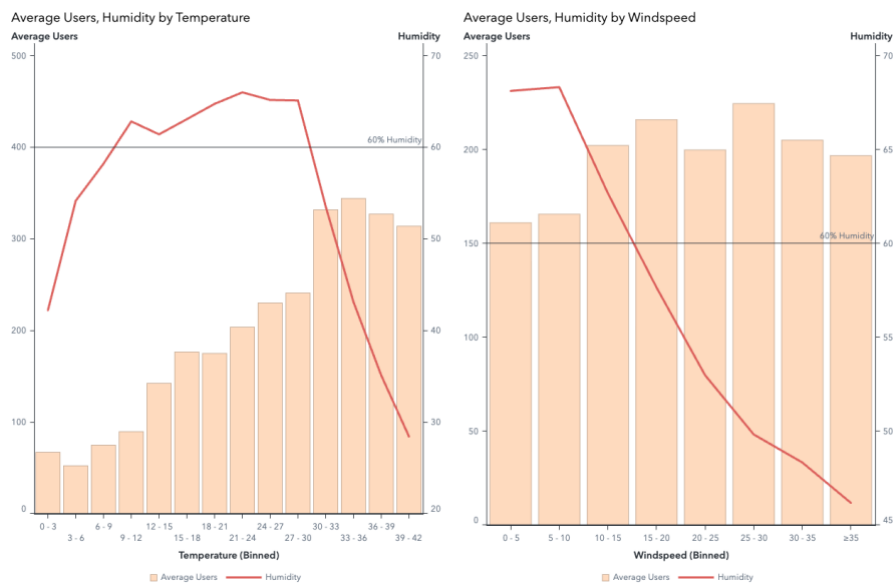


Figure 26: Average Users, Humidity by Temperature/Windspeed

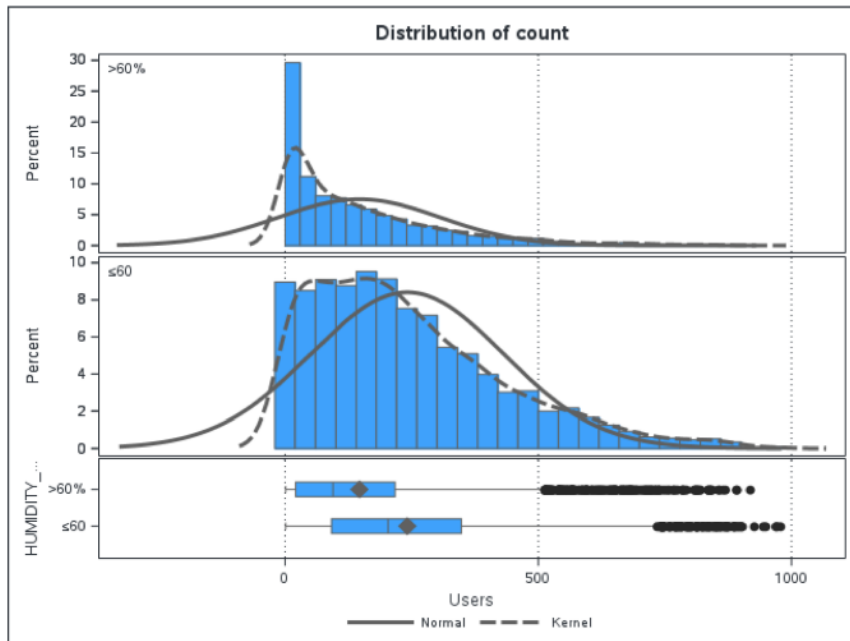


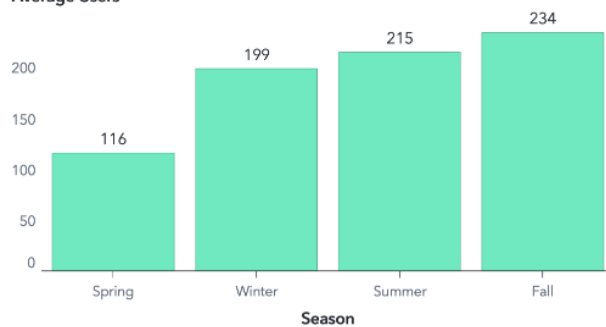
Figure 27: Distribution Count for testing <60% and >60% Humidity

Across the three meteorological factors, humidity shows the clearest monotonic relationship with usage (peak around 20–40% RH followed by steady declines; Welch ANOVA  $p < .0001$ ), while temperature exhibits a comfort-band peak near  $\sim 30\text{--}33^\circ\text{C}$  and windspeed shows a lift from calm to moderate followed by a broad plateau.

Correlation results further indicate a moderate negative coupling between humidity and windspeed and only weak ties with temperature, reinforcing humidity as the most actionable lever and motivating the deeper threshold and sweep analyses.

#### 4.9. Insight 9: Seasonal demand ordering tracks seasonal temperature; humidity and windspeed play secondary roles

Average Users by Season  
Average Users



| Levene's Test for Homogeneity of count Variance<br>ANOVA of Squared Deviations from Group Means |       |                |             |         |        |
|-------------------------------------------------------------------------------------------------|-------|----------------|-------------|---------|--------|
| Source                                                                                          | DF    | Sum of Squares | Mean Square | F Value | Pr > F |
| season_cat                                                                                      | 3     | 8.924E11       | 2.975E11    | 97.58   | <.0001 |
| Error                                                                                           | 10882 | 3.317E13       | 3.0483E9    |         |        |

| Welch's ANOVA for count |        |         |        |
|-------------------------|--------|---------|--------|
| Source                  | DF     | F Value | Pr > F |
| season_cat              | 3.0000 | 335.09  | <.0001 |
| Error                   | 5945.6 |         |        |

| Level of season_cat | N    | count      |            |
|---------------------|------|------------|------------|
|                     |      | Mean       | Std Dev    |
| Fall                | 2733 | 234.417124 | 197.151001 |
| Spring              | 2686 | 116.343261 | 125.273974 |
| Summer              | 2733 | 215.251372 | 192.007843 |
| Winter              | 2734 | 198.988296 | 177.622409 |

Figure 28: Users by Seasons, with ANOVA Statistics

**H<sub>0</sub>:** Mean hourly users are equal across seasons ( $\mu_{\text{Fall}} = \mu_{\text{Summer}} = \mu_{\text{Winter}} = \mu_{\text{Spring}}$ ).

**H<sub>1</sub>:** At least one seasonal mean differs.

Welch's ANOVA shows a clear season effect on users,  $F \approx 335.09$ ,  $p < .0001$ , with means Fall  $\approx 234$ , Summer  $\approx 215$ , Winter  $\approx 199$ , Spring  $\approx 116$ ; Tukey–Kramer indicates all pairs differ ( $R^2 \approx 0.061$ ). This ordering aligns with strong seasonal shifts in temperature—Fall  $\approx 28.8^\circ\text{C}$ , Summer  $\approx 22.8^\circ\text{C}$ , Winter  $\approx 16.6^\circ\text{C}$ , Spring  $\approx 12.5^\circ\text{C}$ —where temperature differences are large (Welch  $F \approx 6874.04$ ,  $p < .0001$ ).

#### 4.10. Insight 10: Meteorology by Season: Big temperature shifts, modest humidity changes, small wind differences

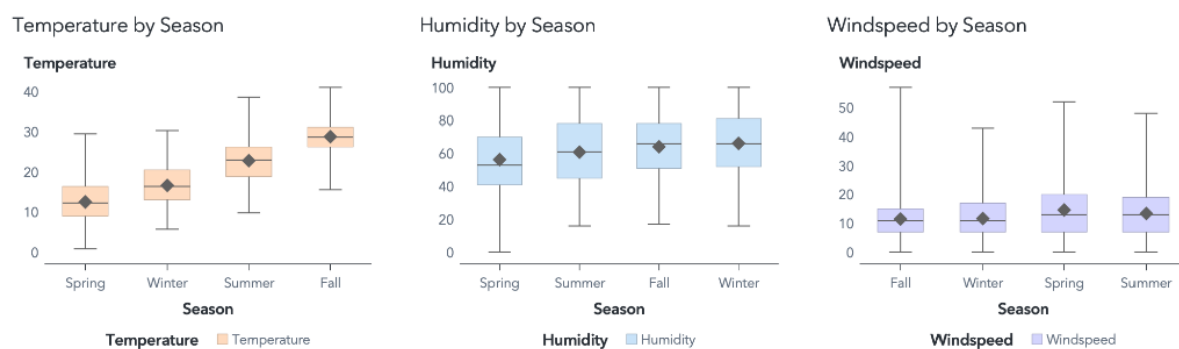


Figure 29: Meteorology by Season box plots

**H<sub>0</sub>:** Seasonal means are equal for each meteorological variable (temperature / humidity / windspeed).

**H<sub>1</sub>:** At least one seasonal mean differs.

Seasonality is also present in humidity (Welch  $F \approx 137.34$ ,  $p < .0001$ ; Winter  $\approx 66\%$ , Fall  $\approx 64\%$ , Summer  $\approx 61\%$ , Spring  $\approx 56\%$ ;  $R^2 \approx 0.037$ ) and in windspeed (Welch  $F \approx 87.35$ ,  $p < .0001$ ; Spring highest  $\approx 14.64$ , Fall/Winter calmest  $\approx 11.5$ – $11.7$ ;  $R^2 \approx 0.025$ ), but these effects are smaller than temperature's. Taken together, the seasonal demand pattern mirrors the temperature profile most closely, with humidity and windspeed providing secondary modulation.

#### 4.11. Insight 11: Small weekend lead, higher single-hour peak on weekdays

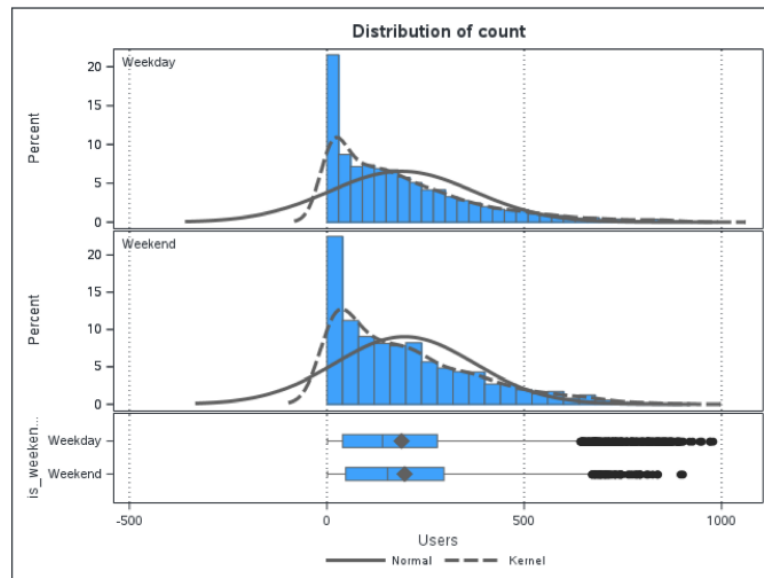


Figure 30: Distribution count of Weekday and Weekend Users

$H_0$ : Mean users (weekday) = mean users (weekend).

$H_1$ : Means differ.

A two-sample comparison was run in SAS Viya with Users (count) as the response and is\_weekend as the grouping factor (weekday vs weekend). Weekend averages are slightly higher than weekdays (197.2 vs 189.3 users/hr). Variances differ (Folded-F  $p \approx 0.027$ ), so Satterthwaite is primary:  $t \approx -2.10$ ,  $p = 0.036$ , 95% CI for (weekday-weekend)  $[-15.37, -0.52]$ ; Wilcoxon corroborates ( $Z \approx 3.52$ ,  $p = 0.0002$ ). Despite the small mean gap, the peak expected hour is higher on weekdays (977 vs 900), implying higher topside provisioning on workdays.

#### 4.12. Insight 12: Hour-matched comparison has no overall difference

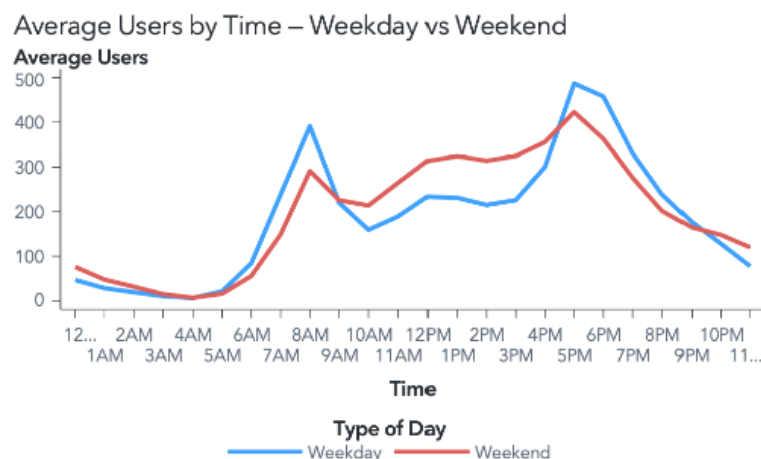


Figure 31: Average Users by Hour, Weekday vs Weekend

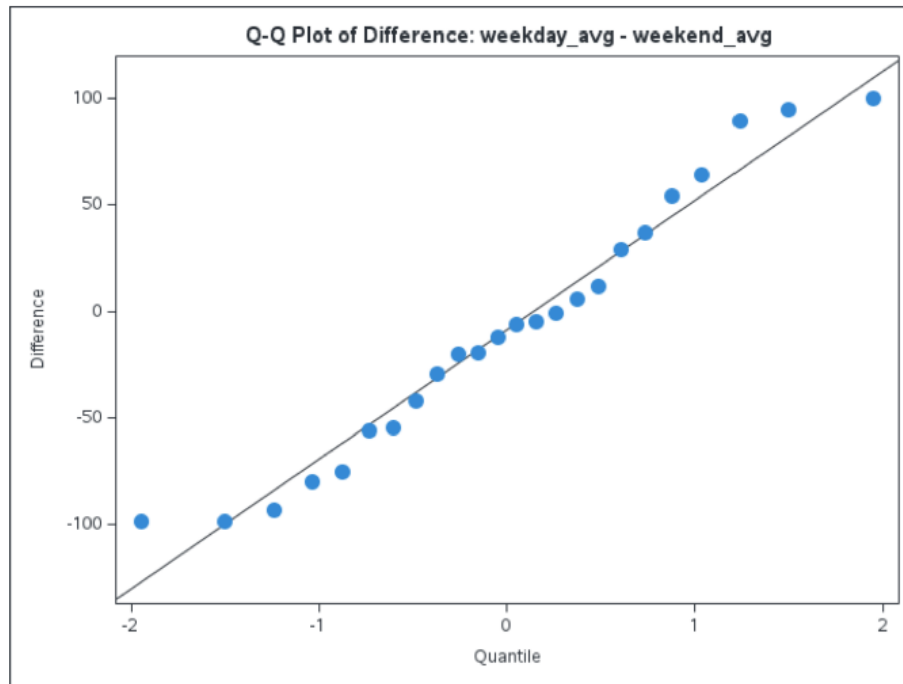


Figure 32: Paired difference if weekend and weekday averages

Variable: \_Difference\_ (Difference: weekday\_avg - weekend\_avg)

| Tests for Normality |           |          |           |         |
|---------------------|-----------|----------|-----------|---------|
| Test                | Statistic |          | p Value   |         |
| Shapiro-Wilk        | W         | 0.955999 | Pr < W    | 0.3634  |
| Kolmogorov-Smirnov  | D         | 0.076724 | Pr > D    | >0.1500 |
| Cramer-von Mises    | W-Sq      | 0.029271 | Pr > W-Sq | >0.2500 |
| Anderson-Darling    | A-Sq      | 0.257798 | Pr > A-Sq | >0.2500 |

Difference: weekday\_avg - weekend\_avg

| N  | Mean    | Std Dev | Std Err | Minimum  | Maximum |
|----|---------|---------|---------|----------|---------|
| 24 | -8.5102 | 60.6659 | 12.3834 | -98.7686 | 100.2   |

| Mean    | 95% CL Mean | Std Dev | 95% CL Std Dev |
|---------|-------------|---------|----------------|
| -8.5102 | -34.1271    | 17.1068 | 60.6659        |
|         |             |         | 47.1504        |
|         |             |         | 85.0997        |

Figure 33: Statistics from Paired Test

$H_0$ : Mean paired difference (weekday-weekend) = 0.

$H_1$ : Mean paired difference  $\neq$  0.

A paired t-test, normality is checked on the paired differences. Across 24 matched hours, the mean difference is  $-8.51$  users/hr with  $t(23)=-0.69$ ,  $p=0.499$ , 95% CI  $[-34.13, 17.11]$  which is not significant. The profile explains this: weekdays concentrate demand into sharp AM/PM commute spikes, while weekends lift later and sustain a broader mid-afternoon plateau, offsetting in the paired average.

#### 4.13. Insight 13: Weekday vs Weekend – Registered Users Lead

```

1  /* Hourly averages of Registered & Casual, by day type */
2  ⊖ proc summary data=CASUSER.HAPPYRIDES_V9 nway;
3      class hour_order is_weekend_cat;
4      var registered casual;
5      output out=WORK._avg_by_hour_type(drop=_type_ _freq_)
6          mean=registered_avg casual_avg;
7  run;
8
9  /* Split into Weekday and Weekend 24-row tables (in WORK) */
10 ⊖ data WORK.regcas_pairs_weekday(keep=hour_order registered_avg casual_avg);
11     set WORK._avg_by_hour_type;
12     if is_weekend_cat='Weekday';
13 run;
14
15 ⊖ data WORK.regcas_pairs_weekend(keep=hour_order registered_avg casual_avg);
16     set WORK._avg_by_hour_type;
17     if is_weekend_cat='Weekend';
18 run;
19
20 /* Copy to CAS for the Task UI */
21 data CASUSER.regcas_pairs_weekday; set WORK.regcas_pairs_weekday; run;
22 data CASUSER.regcas_pairs_weekend; set WORK.regcas_pairs_weekend; run;
23

```

Figure 34: Data Prep for the paired t Test

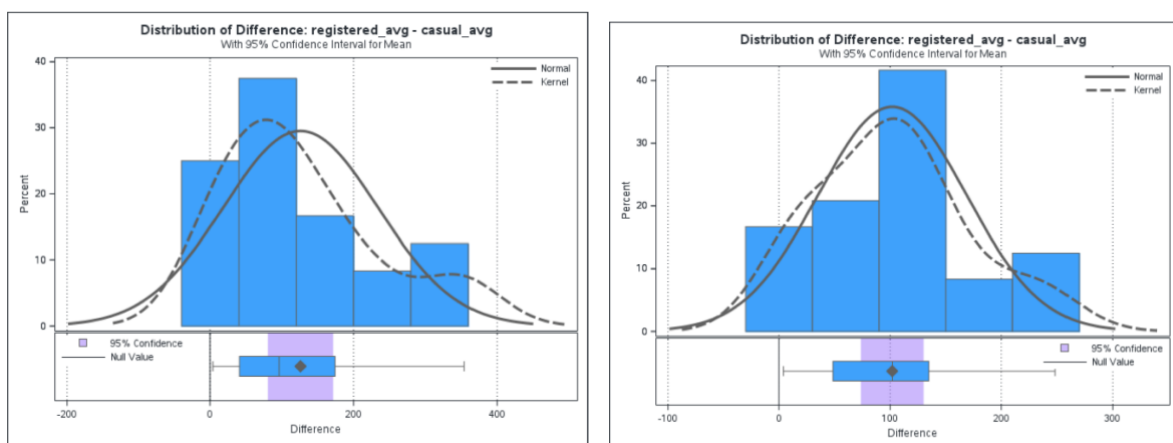


Figure 35: Registered vs Users, Weekend and Weekday Distribution

**Weekdays:** Normality of the paired differences is adequate (Shapiro–Wilk  $W \approx 0.95$ ,  $p \approx 0.30$ ). The mean hourly gap is  $\approx 101.8$  users/hr;  $t(23) = 7.46$ ,  $p < .0001$  with corroborating sign and signed-rank tests (both  $p < .0001$ ). Interpretation: on weekdays, the registered segment dominates across the day and drives the pronounced AM/PM commute spikes seen in the lines.

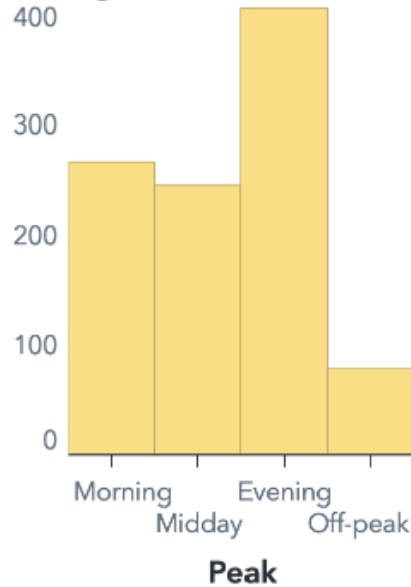
**Weekends:** Differences are somewhat non-normal ( $W \approx 0.88$ ,  $p \approx 0.007$ ), so rely on the t-test alongside nonparametrics. The mean hourly gap is  $\approx 125.9$  users/hr;  $t(23) = 5.70$ ,  $p < .0001$  with sign and signed-rank tests also  $p < .0001$ . Even as casual share expands on weekends (especially mid-afternoon to early evening), registered usage remains higher at most hours, yielding a larger average gap than on weekdays.



#### 4.14. Insight 14: Average Users Dominates Evenings, Morning $\approx$ Midday

### Average Users by Peak

#### Average Users



| Levene's Test for Homogeneity of count Variance<br>ANOVA of Squared Deviations from Group Means |       |                |             |         |        |
|-------------------------------------------------------------------------------------------------|-------|----------------|-------------|---------|--------|
| Source                                                                                          | DF    | Sum of Squares | Mean Square | F Value | Pr > F |
| PEAK_BUCKET                                                                                     | 3     | 1.422E12       | 4.739E11    | 289.75  | <.0001 |
| Error                                                                                           | 10882 | 1.78E13        | 1.6356E9    |         |        |

| Welch's ANOVA for count |        |         |        |
|-------------------------|--------|---------|--------|
| Source                  | DF     | F Value | Pr > F |
| PEAK_BUCKET             | 3.0000 | 1699.41 | <.0001 |
| Error                   | 4010.0 |         |        |

| Level of<br>PEAK_BUCKET | N    | count      |            |
|-------------------------|------|------------|------------|
|                         |      | Mean       | Std Dev    |
| Evening                 | 1824 | 360.855263 | 207.638032 |
| Midday                  | 2279 | 244.557262 | 143.641858 |
| Morning                 | 1820 | 243.189560 | 171.783670 |
| Off-pea                 | 4963 | 86.102156  | 109.201552 |

Figure 36: Average Users by Peaks, and Statistics for Anova Tests

#### Average Users by Day & Time

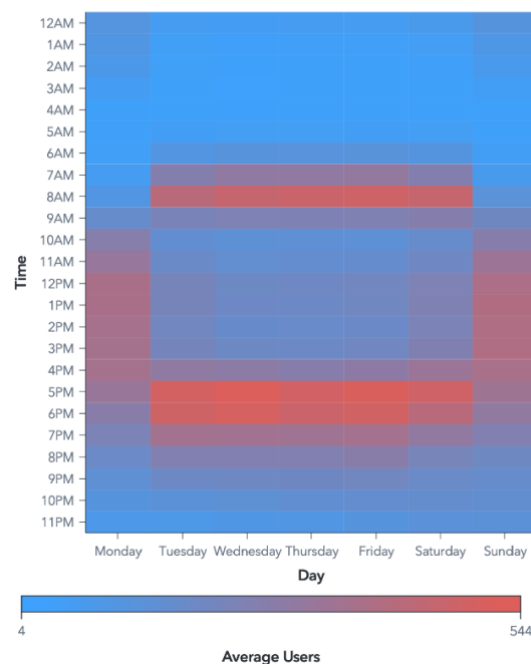


Figure 37: Heatmap of Average Users Day and Time

$H_0$ : Mean users are equal across peak buckets.

$H_1$ : At least one bucket mean differs.

Average users differ markedly across peak buckets: Evening averages 360.9 users/hr and exceeds all other buckets, while Midday (244.6) and Morning (243.2) are

statistically indistinguishable and Off-peak is lowest (86.1). Levene indicated unequal variances ( $p < .0001$ ), so Welch's ANOVA provides the primary inference ( $F = 1699.41$ ,  $p < .0001$ ), with a similar classical result ( $F = 1806.08$ ,  $p < .0001$ ) and model fit  $R^2 \approx 0.332$ ; Tukey–Kramer confirms Evening > Midday and Morning, with Morning  $\approx$  Midday ( $p = 0.9912$ ). These results formalize the heatmap's commute bands: a weekday morning peak around 7–10 AM and a stronger, broader evening peak around 4–8 PM, while weekends shift the morning lift later (~9–11 AM) and extend the evening window (~5–9 PM).

#### 4.15. Insight 15: More users on weekday mornings, no clear weekday and weekend gap in the afternoon.

```

1  /* Create AM-only (07–10) and PM-only (17–20) matched tables */
2  ⊖ data CASUSER.hour_pairs_AM;
3    set CASUSER.hour_pairs_24;
4    if 7 <= hour_order <= 10;
5  run;
6
7  ⊖ data CASUSER.hour_pairs_PM;
8    set CASUSER.hour_pairs_24;
9    if 17 <= hour_order <= 20;
10 run;

```

Figure 38: Data prep for morning vs afternoon/evening on weekday weekend. (Assign1/3\_Confirmatory Data Analysis/9A\_PeakShiftPrep.sas)

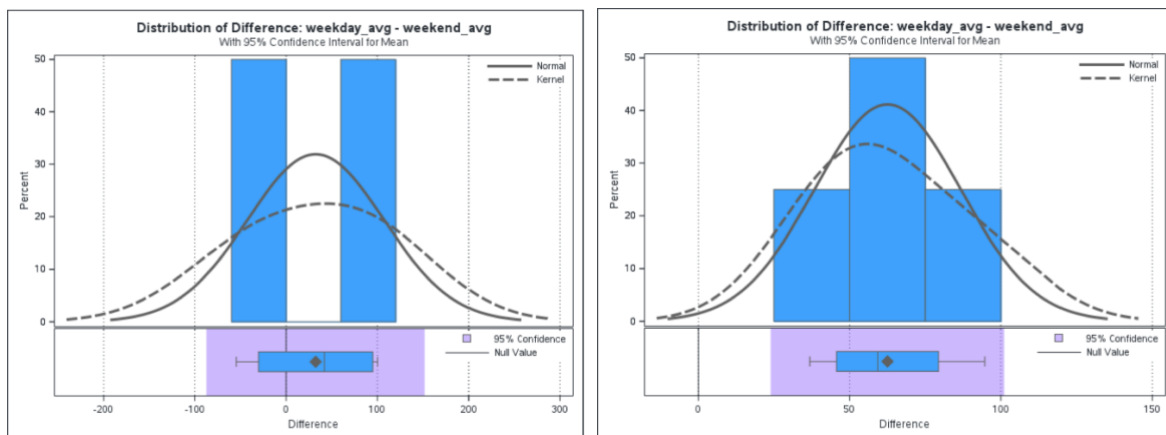


Figure 39: AM/PM vs Weekday Average - Weekend - Average

Variable: \_Difference\_ (Difference: weekday\_avg - weekend\_avg)

| Tests for Normality |           |          |           |         |
|---------------------|-----------|----------|-----------|---------|
| Test                | Statistic |          | p Value   |         |
| Shapiro-Wilk        | W         | 0.884408 | Pr < W    | 0.3578  |
| Kolmogorov-Smirnov  | D         | 0.277276 | Pr > D    | >0.1500 |
| Cramer-von Mises    | W-Sq      | 0.052243 | Pr > W-Sq | >0.2500 |
| Anderson-Darling    | A-Sq      | 0.318312 | Pr > A-Sq | >0.2500 |

Variable: \_Difference\_ (Difference: weekday\_avg - weekend\_avg)

| Tests for Normality |           |          |           |         |
|---------------------|-----------|----------|-----------|---------|
| Test                | Statistic |          | p Value   |         |
| Shapiro-Wilk        | W         | 0.971201 | Pr < W    | 0.8489  |
| Kolmogorov-Smirnov  | D         | 0.22384  | Pr > D    | >0.1500 |
| Cramer-von Mises    | W-Sq      | 0.032082 | Pr > W-Sq | >0.2500 |
| Anderson-Darling    | A-Sq      | 0.208017 | Pr > A-Sq | >0.2500 |

Difference: weekday\_avg - weekend\_avg

| N | Mean    | Std Dev | Std Err | Minimum  | Maximum |
|---|---------|---------|---------|----------|---------|
| 4 | 32.3792 | 75.0503 | 37.5251 | -54.5246 | 100.2   |

| Mean    | 95% CL Mean    | Std Dev | 95% CL Std Dev |
|---------|----------------|---------|----------------|
| 32.3792 | -87.0425 151.8 | 75.0503 | 42.5152 279.8  |

| DF | t Value | Pr >  t |
|----|---------|---------|
| 3  | 0.86    | 0.4517  |

Difference: weekday\_avg - weekend\_avg

| N | Mean    | Std Dev | Std Err | Minimum | Maximum |
|---|---------|---------|---------|---------|---------|
| 4 | 62.5002 | 24.2363 | 12.1181 | 36.8015 | 94.6620 |

| Mean    | 95% CL Mean   | Std Dev | 95% CL Std Dev  |
|---------|---------------|---------|-----------------|
| 62.5002 | 23.9349 101.1 | 24.2363 | 13.7296 90.3661 |

| DF | t Value | Pr >  t |
|----|---------|---------|
| 3  | 5.16    | 0.0141  |

Figure 40: Statistics for AM/PM, Weekday - Weekend Average

$H_0$ : Mean paired difference = 0.

$H_1$ : Mean paired difference  $\neq$  0.

Paired test was conducted to examine the differences. In the AM window (four matched hours), the weekday–weekend gap is sizeable and statistically reliable: mean difference  $\approx$  62.5 users/hr,  $t(3)=5.16$ ,  $p=0.014$ , indicating weekday AM > weekend AM even with only four pairs. The corresponding lines show sharp weekday surges around 7–10 AM, consistent with commute-driven demand. In the PM window (four matched hours), the mean difference is  $\approx$  32.4 users/hr,  $t(3)=0.86$ ,  $p=0.452$ , which is not significant; with such a small  $n$ , the test has low power and the confidence interval is wide, so a true modest difference cannot be ruled out. The shape of the curves helps explain the contrast: weekdays exhibit peaky PM usage centered near 4–8 PM, whereas weekends sustain a broader evening plateau ( $\approx$  5–9 PM), flattening the hour-matched difference. Taken together with the peak-bucket ANOVA, the evidence points to commute-specific weekday intensity in the morning and broader, later weekend activity in the evening.

## 5. Interpretation of Results: What Drives HappyRides?

Humidity emerges as the dominant weather constraint on demand. The observed monotonic drop in usage at higher relative humidity and the presence of a practical discomfort threshold align with multi-city evidence that humidity suppresses cycling by increasing thermal strain and perceived exertion, often operating in threshold-like rather than strictly linear ways. Recent reviews and case studies describe humidity as a primary deterrent, frequently interacting with other weather to compound discomfort. (Cortez-Ordonez, Vazquez, & Sanches-Espigares, 2024)

Temperature influences demand in a non-linear “comfort-band” fashion. Usage typically rises from cool conditions into a pleasant range and then softens as heat builds, an inverted-U pattern that has been documented for bikeshare and commuter cycling across multiple cities using flexible models. and longitudinal analyses. This shape is consistent with thermoregulatory comfort: modest warmth supports riding, while heat stress reverses gains beyond the comfort zone. (Heaney, Carrion, Burkart, Lesk, & Jack, 2019)

Windspeed plays a secondary, context-dependent role. The data indicate that moving out of very calm conditions is associated with improved usage, after which differences flatten across mid-range winds, a plateau pattern that mirrors findings that wind effects are generally smaller than temperature and humidity, with pronounced deterrence mostly at higher gusts or in combination with other adverse weather. Studies comparing elasticities across variables typically rank wind below temperature and humidity in explanatory power for bikeshare volumes.

Seasonality in demand largely tracks seasonal temperature, with humidity and wind providing modulation rather than wholesale re-ordering. The literature consistently reports higher ridership in warmer seasons and reductions in colder or wetter periods, with temperature shouldering most of the seasonal signal and other meteorological factors acting as amplifiers or dampeners around that baseline. (Wang & Noland, 2021)

Temporal structure explains when usage materializes once meteorology sets the baseline. Weekdays concentrate demand in commute bands, whereas weekends shift activity later and broaden evening use; member/registered riders anchor the weekday peaks, while casual riders expand in leisure windows. These profiles match cross-system observations that commuter-oriented segments dominate weekday peaks and that leisure-oriented segments inflate weekend totals without overturning member dominance. (Yang, Brakewood, Nicolas, & Sion, 2019)

## 6. Managerial Communication

The analysis assessed meteorological factors, temporal structure, and rider segments. A more honest and complete analysis would be possible by incorporating the following enhancements:

### 6.1. Hourly Completeness

Ensuring hourly completeness on a canonical clock is foundational. The table should contain exactly 24 rows per service day (00:00–23:00). This prevents hidden bias from storm hours or overnight gaps and stabilizes all commute-window metrics and averages. With a consistent time base, comparisons by hour, weekday/weekend, and seasonal windows reflect true behavior rather than data collection artifacts.

### 6.2. Precipitation

Precipitation magnitude and duration are among the strongest short-horizon demand suppressors, add continuous fields for intensity and exposure time while retaining the human-readable weather category. Distinguishing mist/drizzle from sustained or heavy rain meaningfully improves fit and makes hour-ahead planning more actionable (e.g., differentiating a brief shower from a prolonged downpour during the evening peak).

### 6.3. Thermal-Comfort Indices

Given the dominant role of humidity, derive thermal-comfort indices that jointly capture heat and moisture (e.g., `heat_index_c`). These measures align naturally with the observed threshold behavior (e.g., the 60% RH trigger) and reduce the risk of misattributing humidity-driven discomfort to temperature alone. Including comfort indices alongside raw temperature and humidity supports clearer narratives and more stable segment comparisons.

### 6.4. Event Calendar

Embed calendar and event context directly in the dataset with `is_holiday`, `school_term`, and `major_event` flags at the relevant geography. This structure explains systematic exceptions such as holiday morning softness, school-term commuter surges, or event-driven evening spikes, and increases forecast specificity for particular dates without post-hoc adjustments.

### 6.5. Bike Dock Stations

Finally, incorporate supply and constraint markers to separate true demand changes from operational limits. For station systems, persist `bikes_available` and `docks_available`; for free-floating systems, include a `rebalance_flag` and `service_zone`. These fields prevent weather or season from being blamed when a trough is actually a stockout or a temporary service restriction which is critical for accurate attribution and for sizing interventions.

## 7. References

- Cortez-Ordóñez, A., Vázquez, P.-P., & Sánchez-Espigares, J. A. (2024). Features that influence bike sharing demand. *Heliyon*, Volume 10, Issue 18.
- Heaney, A., Carrion, D., Burkart, K., Lesk, C., & Jack, D. (2019). Climate Change and Physical Activity: Estimated Impacts of Ambient Temperatures on Bikeshare Usage in New York City. *Environ Health Perspect*, doi: 10.1289/EHP4039.
- Wang, H., & Noland, R. (2021). Bikeshare and subway ridership changes during the COVID-19 pandemic in New York City. *Transp Policy*, doi: 10.1016/j.
- Yang, S., Brakewood, C., Nicolas, V., & Sion, J. (2019). Bikeshare Users on a Budget? Trip Chaining Analysis of Bikeshare User Groups in Chicago. *Sagepub*, DOI: 10.1177/0361198.
- Eren, E., & Emre Uz, V. (2020). A review on bike-sharing: The factors affecting bike-sharing demand. *Sustainable Cities and Society*.